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AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors: US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - Chapter IV

**78.9% global operational AI
compute = USA**

1.59x energy cost EU/US

**3.64:1 CACI ratio US/EU (Power
Mode)**

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CHAPTER IV

Mechanisms of American Competitive Advantage Through AI

Chapter III established the facts: the United States concentrates 78.9 percent of operational global AI compute (49.9 percent if planned clusters are included), 45 percent of data center energy consumption, and controls 70 percent of the European cloud market through three hyperscalers. This chapter analyses the mechanisms through which this compute asymmetry translates into measurable competitive advantage. We identify four transmission channels: training costs as a barrier to entry, cloud concentration as a vector of dependence, regionally differentiated productivity, and the capture of innovation rents by first movers.

4.1 Training Costs as a Barrier to Entry

4.1.1 The Exponential Explosion of Training Costs

The most documented channel is that of foundation model training costs. Martens (Bruegel, October 2024) establishes that the cost of training a frontier model rose from 1,000 USD in 2017 to nearly 200 million USD in 2024, and could reach several billion by 2030.[1] This exponential growth follows the scaling laws identified by Kaplan et al. (2020): the improvement of cognitive performance in models is subject to constant returns in complementary inputs (parameters, training data, compute capacity). In other words, each marginal gain in performance requires a proportional increase in compute, data, and model size.

Cottier et al. (2024), cited by Martens, decompose training costs into five components: personnel, AI chips, server costs, network interconnection, and energy. They estimate infrastructure costs (hardware plus data center) at ten times the cost of training itself, with a depreciation rate of 140 percent per year (complete depreciation in 8.5 months, matching the renewal pace of chip generations). GPT-4 infrastructure by end-2023 was estimated at approximately 800 million USD. By extrapolation, infrastructure costs could reach 500 billion USD by 2030, replicated across half a dozen hyperscalers.[2]

4.1.2 The US Advantage: Abundant and Subsidised Compute

In this context, access to cutting-edge compute becomes the discriminating factor. American Big Tech (Microsoft, Meta, Google, Amazon, xAI) collectively invested 320 billion USD in 2025 in AI infrastructure (servers, data centers, chips), up from 230 billion USD in 2024.[3] Meta is developing an AI computing system with 350,000 H100 processors, estimated at over 10 billion USD. As Martens notes, this barrier to entry is completely beyond the reach of public funding; only the largest companies can aspire to it.[4]

For European actors, this asymmetry translates concretely. The cost of training a model comparable to GPT-4 is estimated at approximately 100 million USD in the United States (with direct GPU access, cheap energy, amortisation over a massive installed base). In Europe, the same training would cost 2 to 5 times more, due to: (i) indirect compute access via US cloud (hyperscaler margins), (ii) energy costs that run 1.4 to 1.7 times higher after PPA correction (and 2 to 3 times higher on unadjusted Eurostat industrial tariffs - Chapter III), and (iii) the absence of comparable economies of scale. The FLOP cost gap (CACI metric M2) is estimated between 2.4x and 3.6x once cloud markups, energy and amortisation are stacked, even though the underlying PPA-adjusted energy ratio is only 1.59x.[5]

4.1.3 Forced Co-opetition: European Startups and US Big Tech

Exponential costs create a mechanism of structural dependence. As analysed by Martens (Bruegel), AI startups - including European ones - are forced to cooperate with American Big Tech to access computing infrastructure and distribution channels. This co-opetition (forced cooperation with a competitor) places startups in a subordinate position: they depend on US platforms to train their models, then find themselves in direct competition with the proprietary models of those same platforms. The UK Competition and Markets Authority (CMA, 2024) and the European Commission have identified these agreements as potentially anti-competitive.[6] Azoulay et al. (2024), cited by Martens, argue that resource sharing is the only path to avoid concentration, unless a firm actor uses openness as a competitive strategy, which Meta has partially done with LLaMA.

4.2 Cloud Concentration as a Vector of Geopolitical Dependence

4.2.1 The European Cloud Market: 70 Percent Under US Control

The second mechanism is the concentration of the European cloud market. Synergy Research Group (July 2025) establishes that the European cloud infrastructure services market reached 61 billion EUR in 2024 (multiplied by six since 2017) and should exceed 75 billion EUR in 2025 (+24 percent). Amazon (AWS),

Microsoft (Azure), and Google (GCP) account for 70 percent of this market. The share of European providers dropped from 29 percent in 2017 to 15 percent in 2022, where it has stagnated since.[7]

Among European providers, SAP and Deutsche Telekom lead with approximately 2 percent market share each, followed by OVHcloud, Telecom Italia, and Orange. John Dinsdale, Synergy chief analyst, observes that US providers invest 10 billion EUR per quarter in European capex, and that the three hyperscalers now operate more than 140 hyperscale data centers in Europe.[8] The fastest growth (140 to 160 percent) concerns services specifically related to generative AI (GPU-as-a-Service, GenAI PaaS), a segment where European players are virtually absent.

4.2.2 From Commercial Lock-in to Geopolitical Lock-in

This concentration goes beyond a simple commercial issue. The concept of geopolitical vendor lock-in (defined in Chapter I) describes a situation where technical dependence on a foreign provider is compounded by a regulatory risk linked to the sovereign decisions of that provider home country. The CLOUD Act (2018) requires American cloud providers to furnish US authorities with data hosted on their servers, including those located in Europe. In May 2025, Microsoft France chief legal officer acknowledged under oath the inability to guarantee that French citizens data would never be transmitted to American authorities.[9]

In the AI context, geopolitical lock-in operates at three levels. At the infrastructure level (IaaS), AI workloads are deployed on servers physically controlled by entities subject to US jurisdiction. At the platform level (PaaS), training and inference frameworks are integrated into proprietary ecosystems (Azure ML, SageMaker, Vertex AI). At the model level (MaaS - Model-as-a-Service), access to state-of-the-art models (GPT-4, Claude, Gemini) goes through APIs controlled by US companies, with terms of service that can be modified unilaterally. Section 232 protectionism, by differentiating the cost of hardware access, reinforces all three levels simultaneously.

The Chapter I distinction between Physical CACI (F_{total}) and Sovereign CACI (F_{dom}) is critical here: while EU operational compute is largely owned by EU operators (OVH, Scaleway, Sesterce, EuroHPC, AI Factories), EU AI workloads run predominantly on US-owned infrastructure. The dependence is not on installed compute, but on used compute, and on the platforms and models layered above it.

4.2.3 The Draghi Report and the European Diagnosis

The Draghi Report (September 2024), commissioned by the European Commission, constitutes the most comprehensive institutional recognition of this dependence. Draghi observes that European restrictions on data storage and processing create high compliance costs and hinder the creation of large integrated datasets for AI model training. He recommends the creation of a European hyperscale cloud infrastructure, the consolidation of European cloud providers, and massive investment in AI, quantum computing, and EuroHPC supercomputers.[10] Martens (Bruegel, 2025) moderates this optimism, however, observing that EuroHPC supercomputers are not designed for generative AI and that upgrading to competitive standards exceeds the financial capacity of European budgets.[11]

4.3 Differentiated Productivity: The Abundant Compute Dividend

4.3.1 The Theoretical Macroeconomic Potential

The third transmission channel is AI impact on labor productivity and its regional asymmetry. The McKinsey Global Institute (May 2024, December 2025) estimates that generative AI could enable Europe to reach an annual productivity growth rate of 3 percent per year by 2030 in an accelerated adoption scenario with proactive workforce redeployment.[12] This figure is sufficient to close most of the productivity gap with the United States. However, in a slow adoption scenario, productivity growth would be only 0.3 percent, close to current levels and insufficient to fund the European social model.

The IMF (Working Paper, March 2025) provides a more granular analysis.[13] By cross-referencing task exposure indices to AI (Felten et al., Eloundou et al.) with sectoral data, the IMF estimates total factor productivity (TFP) gains significantly higher in advanced economies than in middle-income countries, due to their greater share of value added in high AI-exposure sectors (financial services, consulting, technology) and their higher wage levels, which further justify automation. Underlying microeconomic studies (randomized controlled trials) show gains ranging from 14 percent for low-skilled tasks to over 50 percent for software engineers.

4.3.2 The Gap Conditioned by Compute Access

Yet, the realisation of these productivity gains is conditioned by compute access. This is where the asymmetry diagnosed in Chapter III produces its effects. The French AI Commission (2024) estimates AI potential impact on French growth at 1.3 percentage points per year, by analogy with the effects of electricity, but this estimate assumes unconstrained access to cutting-edge compute. Similarly, McKinsey (December 2025) conditions the optimistic European scenario on accelerated adoption which implies massive infrastructure investments.[14]

The investment gap is considerable. McKinsey (January 2026) estimates that large European companies face an annual deficit of 700 billion USD in R&D and capital expenditure compared to their American counterparts. In digital technologies alone, US companies invested 2 trillion EUR more than European companies over the 2021-2025 period.[15] The annualised gap between January 2024 and September 2025 amounts to 580 billion EUR for corporate investment and 300 billion EUR for startups and scale-ups.

This investment deficit, combined with the compute asymmetry and energy costs, produces a conditional AI productivity gap. In our modelling, we distinguish between theoretical potential (identical for US and EU with comparable sectoral structures) and achievable potential, which depends on effective compute access. With a CACI(US)/CACI(EU) of 3.64:1 in Power Mode (Chapter III) on the April 2026 dashboard snapshot, the achievable European potential is structurally inferior. Translated into normalised CACI scores, the EU sits at 28.9 against a US benchmark of 100, France at 25.3, Germany at 5.4 - the gap is therefore not 7 to 12 times inferior as earlier analyses suggested, but is precisely 3.64 to 1 in compute-adjusted competitive intensity, with sharp variation across member states.

4.4 Innovation Rent Capture: A Self-Reinforcing Advantage

4.4.1 The First-Mover Advantage Mechanism in AI

The fourth channel is the most structurally significant in the long run. The theory of general purpose technologies (GPT, Bresnahan and Trajtenberg, 1995) predicts that early adopters of a general technology capture disproportionate innovation rents, which become self-reinforcing through three mechanisms.

First, data scale effects. Companies that deploy AI first accumulate usage data (feedback loops, fine-tuning) that improves the performance of their models. This data is non-rival but exclusive: once captured by an actor, it does not benefit competitors. McKinsey estimates that 75 percent of generative AI economic potential is concentrated in four key functions (customer service, software development, marketing, R&D), where usage data is particularly determinative.[16]

Second, platform network effects. US hyperscalers benefit from a virtuous circle: more clients leads to more revenue, which leads to more infrastructure investment, which leads to a better offering, which leads to more clients. This mechanism explains why, despite discussions on digital sovereignty, European providers share stagnates at 15 percent: the service quality gap (latency, range of services, technical support, integration ecosystem) is too large to be bridged by sovereignty arguments alone.

Third, talent capture. US companies offer compensation packages including salaries and equity that attract the best AI researchers and engineers worldwide, including Europeans. Martens notes that qualified AI personnel are scarce and that companies compete fiercely to recruit them, with compensation packages difficult for European institutions to match. This AI brain drain weakens the European CACI L(r) factor and reinforces the American advantage in human capital.[17]

4.4.2 The Risk of Irreversibility

These three mechanisms produce a potentially irreversible cumulative effect. As the OECD analyses (Martens, 2021) by analogy with previous technological waves (broadband internet, cloud), latecomers in the adoption of a general technology become structurally dependent on foreign platforms, suffer compressed margins, and progressively lose their strategic autonomy. The 2022 semiconductor shortage, which cost the European automotive sector alone 100 billion EUR, illustrates the concrete consequences of unanticipated technological dependence.[18]

The trajectory is all the more concerning as the EU is absent from four of the eight segments of the generative AI value chain as mapped by McKinsey (2024): AI-design semiconductors (dominated by Nvidia), AI cloud platforms (AWS, Azure, GCP), foundation models (OpenAI, Google, Anthropic, Meta), and AI development tools. It is competitive only in downstream application segments: sectoral applications (SAP, Siemens), industrial integration, and specialised semiconductors (power semiconductors: Infineon, STMicroelectronics, NXP, with about 15 percent global market share).[19]

4.5 Synthesis: The Mechanics of US Competitive Advantage

The four channels identified form a reinforcing system. The compute asymmetry (Chapter III) feeds differentiated training costs (4.1), which strengthen dependence on US cloud (4.2), which constrains achievable productivity (4.3), which slows European investment and reinforces rent capture by American

players (4.4), which in turn widens the compute gap. With the April 2026 snapshot of 78.9 percent operational compute share, 1.59x PPA-adjusted energy cost and a CACI Power Mode ratio of 3.64:1, this circle constitutes what Farrell and Newman (2019) would describe as structural interdependence weaponisation: European dependence on American infrastructure, initially founded on economic efficiency, becomes a lever of geopolitical power when instrumentalised by measures such as Section 232.

Trump protectionism (Chapter III, phase 4) did not create this asymmetry; it pre-existed massively. It institutionalises and legalises it through a tariff mechanism that differentiates the cost of compute access by nationality. In doing so, it transforms a de facto advantage into a de jure advantage, whose dismantling is politically and legally far more difficult. Chapter V examines how these mechanisms could evolve across four prospective scenarios.

Table 8. European cloud market: US hyperscaler domination.

Source: Synergy Research Group (July 2025).

Indicator	2017	2022	2024	Trend
EU Cloud Market (billion EUR)	~10	~35	61	x6 in 7 years
AWS+Azure+GCP Share	~50 pct	~67 pct	70 pct	Growing
EU Provider Share	29 pct	15 pct	15 pct	Stable (floor)
US Capex in Europe	n.a.	n.a.	~40 bn EUR/yr	10 bn EUR/quarter
Hyperscale DCs in EU	~40	~80	>140	+75 pct in 2 years

Table 9. AI productivity and competitive advantage: US vs EU (April 2026 calibration).

Sources: McKinsey (2024, 2025, 2026), Bruegel/Martens (2024), CACI calibration on the public dashboard.

Dimension	United States	Europe (EU)	Gap
Potential AI productivity (McKinsey, accel. scen.)	+2.5-3.5 pct/yr	+2.5-3.0 pct/yr	Comparable
Achievable AI productivity (under compute constraints)	+2.5-3.0 pct/yr	+0.8-1.5 pct/yr	-1.5 to -2 pts
Annual AI investment (corporate + VC)	~450 bn USD	~50-70 bn EUR	~7:1
FLOP cost (USD/TFlop, training)	~0.5	~1.2-1.8	2.4-3.6x
Normalised CACI Power Mode (USA = 100)	100	28.9 (EU(13))	3.64:1

Notes

1. Martens, B. (2024), 'The Tension Between Exploding AI Investment Costs and Slow Productivity Growth,' Working Paper 18/2024, Bruegel, pp. 5-8. The time series is calibrated on Epoch AI training cost data for frontier models.
2. Cottier, B. et al. (2024), cited in Martens (2024), op. cit. The 10x factor between infrastructure and training costs is a central estimate, subject to variation based on post-training infrastructure utilisation.
3. IEA (2025), Energy and AI, Paris, p. 42. The IEA compiles investment announcements from Meta, Amazon, Alphabet, Microsoft and xAI for 2025.

4. Martens, B. (2024), 'Why Artificial Intelligence Is Creating Fundamental Challenges for Competition Policy,' Policy Brief 16/2024, Bruegel. Meta plans a system of 350,000 H100s (at about 30,000 USD per unit), i.e. over 10 billion USD in hardware alone.
5. Author estimate based on Bruegel (training costs), Eurostat (industrial electricity tariffs), EIA (US prices), the public dashboard PPA-adjusted reference values (USA 85, EU 135 USD/MWh), and Epoch AI (GPU performance/price). The 2.4 to 3.6x range stacks cloud markup (1.5 to 2x) on top of the 1.59x PPA-adjusted energy ratio and the absence of EU economies of scale; it depends on access mode (cloud vs on-premise) and PPA negotiations.
6. Competition and Markets Authority (2024), 'AI Foundation Models - Update Report,' London. The European Commission launched a call for contributions on competition in generative AI in 2024.
7. Synergy Research Group (July 2025), 'European Cloud Providers Local Market Share Now Holds Steady at 15 percent.' The market is defined as IaaS plus PaaS plus hosted private cloud.
8. Dinsdale, J. (2025), cited in Synergy Research Group, op. cit.: US cloud providers continue to invest 10 billion EUR every quarter in European capex, most of which comes from the big three.
9. Data Center Dynamics (July 2025), 'European Cloud Providers Hold 15 percent of Local Market Share.' The article reports the testimony of Anton Carniax, Microsoft France chief legal officer, before a parliamentary committee.
10. Draghi, M. (September 2024), The Future of European Competitiveness, report commissioned by the European Commission. Cited in Martens, B. (2025), 'Catch-Up with the US or Prosper Below the Tech Frontier? An EU Artificial Intelligence Strategy,' Bruegel Policy Brief.
11. Martens, B. (2025), op. cit. The author notes that the EuroHPC approach, centred on hardware, has characterised European digital policies for decades and is particularly poorly oriented regarding AI.
12. McKinsey Global Institute (May 2024), 'A New Future of Work: The Race to Deploy AI and Raise Skills in Europe and Beyond.' The 3 percent figure assumes an accelerated adoption scenario with full workforce redeployment.
13. IMF (March 2025), 'Artificial Intelligence and Productivity in Europe,' Working Paper WP/25/067. The analysis uses Felten et al. (2021) and Eloundou et al. (2024) exposure indices.
14. McKinsey (December 2025), 'Accelerating Europe AI Adoption: The Role of Sovereign AI Capabilities.' The report estimates European labor productivity growth at 0.2 percent on average over 2020-2025, versus a potential of 3.1 percent with full AI adoption.
15. McKinsey (January 2026), 'Transforming Europe: Bold Moves to Lift a Continent.' The 700 bn USD per year gap is calculated on the 150 largest technology companies (US vs EU) in capex plus R&D (S&P Capital IQ data).
16. McKinsey Global Institute (2024), 'The Economic Potential of Generative AI: The Next Productivity Frontier.' The four functions (customer service, software development, marketing/sales, R&D) represent about 75 percent of the identified economic potential across 63 use cases.
17. Martens, B. (2024), Working Paper 18/2024, op. cit. Personnel costs (salaries plus equity) are often the most significant component of AI model development costs.
18. McKinsey (October 2024), 'Time to Place Our Bets: Europe AI Opportunity.' The 100 bn EUR GDP loss figure for the European automotive industry comes from Allianz (September 2022) and OICA.
19. McKinsey (October 2024), op. cit. European semiconductors (Infineon, STMicro, NXP) represent about 15 percent of the global integrated design-manufacturing market (Omdia/Informa data, 2024), but primarily in power and automotive segments, not AI GPU/ASIC.

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